# F70LA

## Solutions December 2018

Q1 (a) $p_{63} = \frac{1025-63}{1000} = 0.962$

$\therefore q_{63} = 1 - 0.962$

$= 0.038$

(b) $p_{[58]} = (0.967)^{0.3}$

$= 0.9900$

$p_{[58]+1} = (0.966)^{0.8}$

$= 0.9727$

$p_{60} = 0.965$

$\therefore {}_3p_{[58]} = 0.9900 \times 0.9727 \times 0.965$

$= 0.929259$

(c) P(dies in year 1) $p_{[55]} = 0.9909$

$= 0.00910$

$p_{[55]+1} = (0.969)^{0.8}$

$= 0.9751$

$\therefore$ P(dies in year 2) $= 0.9909\,(1 - 0.9751)$

$= 0.02465$

$\therefore$ P(payment at time 3) $= 1 - 0.00910 - 0.02465$

$= 0.96625$

Q1 (c) continued

| $t$ | $p$ | $v^t$ | $pv^t$ | $p(v^t)^2$ |
|---|---|---|---|---|
| 1 | 0.00910 | 925,926 | 8,422 | $7.798 \times 10^9$ |
| 2 | 0.02465 | 857,339 | 21,135 | $18.119 \times 10^9$ |
| 3 | 0.96625 | 793,882 | 767,042 | $608.902 \times 10^9$ |
| | | | 796,599 | $634,820 \times 10^9$ |

$E(X) = £796,599$

(d) $Var(X) = E(X^2) - [E(X)]^2$

$= 6.3482 \times 10^{11} - (796,599)^2$

$= (£15,827)^2$

$SD = £15,827$

(e) $10^6 v^3 = £793,832$

$E(X)$ is a little higher, because there is a small probability of payment before $t=3$.

(f) Higher.

More uncertainty, because the smaller probabilities are increased.

Q2

(a) $P \ddot{a}^{13.780}_{45:\overline{20}|} = 100{,}000 \, A^{0.46998}_{45:\overline{20}|}$

$P = £3{,}410.60$ pa

(b) $100{,}000 \, A_{52:\overline{13}|} - P \, \ddot{a}_{52:\overline{13}|}$

$= £26{,}663$

or £26,770 if £3400 is used

(c) Value of premiums $= (400 \times 12) \, \ddot{a}^{(12)}_{52}$

$\ddot{a}^{(12)}_{52} = \ddot{a}^{16.838}_{52} - \frac{11}{24}$

$= 16.380$

$\therefore$ value $= £78{,}622$

$\therefore 26{,}663 + 78{,}622 = 200{,}000 \, A^{.35238}_{52} + X (\bar{I}A)^{8.59412}_{52} + 100$

$X = £4{,}050$ $£4038.68$

or £4051.14

(d) Immediate big increase in SA on death
(£100,000 → £204,050)
⇒ suspicious
⇒ Consider medical questionnaire/examination?

Q2 (e) Possibly.

But this may have been compensated for, by using higher mortality or a lower interest rate

(which is not ideal, because we don't know if "higher" is high enough.)

Q3 (a) $S_0(0) = (50-0)/50 = 1$ ✓

$S_0(\lim_{t\to\infty} t) \neq 0$ ×

So no.

(b) $S_0(0) = e^0 = 1$ ✓

$\lim_{t\to\infty} S_0(t) = 0$ ✓

$\frac{d}{dt} S_0(t) = -3e^{-3t} < 0$ ✓

So yes.

(c) $S_0(0) = 700/700 = 1$ ✓

$S_0(70) = 0$ ✓

$\frac{d}{dt} S_0(t) = (60-2t)/700$

$> 0$ for small $t$ ×

So no.

Q4

(a) $13.5\,p_{70.3} = \dfrac{L_{83.8}}{L_{70.3}}$

$$= \frac{L_{71}}{L_{70.3}} \times \frac{L_{83}}{L_{71}} \times \frac{L_{83.8}}{L_{83}}$$

$$= \left(\frac{L_{71}}{L_{70}}\right)^{0.7} \times \frac{L_{83}}{L_{71}} \times \left(\frac{L_{84}}{L_{83}}\right)^{0.8}$$

$L_{71} = 9299.747$, $L_{70} = 9392.621$, $L_{83} = 6826.442$, $L_{71} = 9299.747$, $L_{84} = 6483.987$, $L_{83} = 6826.442$

$$= 0.69955$$

(b) $3.215\,q_{90.8} = \dfrac{L_{94}}{L_{90.8}}\left(1 - \dfrac{L_{99}}{L_{94}}\right)$

$$= \left(\frac{L_{91}}{L_{90}}\right)^{0.82} \left(\frac{L_{94}}{L_{91}}\right) \left(1 - \frac{L_{99}}{L_{94}}\right)$$

$L_{91} = 3706.149$, $L_{90} = 4118.693$, $L_{94} = 2530.033$, $L_{91} = 3706.149$, $L_{99} = 1028.591$, $L_{94} = 2530.033$

$$= \cancel{0.37232}\ 0.39666$$

Q5/

Under UDD, $_sq_x = s q_x$

where $x$ is an integer

and $0 \leq s \leq 1$

$$\mu_{x+s} = \frac{-\frac{d}{ds}\left({}_sp_x\right)}{{}_sp_x}$$

Now $\frac{d}{ds}\left({}_sp_x\right) = \frac{d}{ds}\left(1 - {}_sq_x\right)$

$$= -\frac{d}{ds}\left({}_sq_x\right)$$

$$= -\frac{d}{ds}\left(s q_x\right)$$

$$= -q_x$$

$$\therefore \mu_{x+s} = \frac{q_x}{{}_sp_x}$$

${}_sp_x$ is decreasing with $s$,

so $\mu_{x+s}$ is increasing.

Q6

(a) At the moment of sale,

Net income $= 100 \times £160,000 - £900,000$

$= £15,100,000$ (1)

Value of liabilities $= 100 \times \left(10,000\, a_{65}^{12.666}\right) \times 1.02$

$\left(\ddot{a}_{65} - 1\right)$

$= £12,919,320$ (2)

Profit $= (1) - (2)$

$= \underline{\underline{£2,180,680}}$

(b) - Mortality was lower in premium basis than in valuation basis, or
- Interest rate was lower, or
- Expense loading was higher, or
- There was a loading for profit

[1 mark each, max 2]

(c) - Commission paid for sellers
- or cost of setting up contracts
[but not medical underwriting]

(d) Age 79

Assets at start of year
$= 83 \times (10{,}000\ \overset{6.877}{a_{79}}) \times 1.02$

$\overset{6.877}{a_{79}} = (\ddot{a}_{79} - 1)$

$= £5{,}822{,}068$

At end, assets $= (5{,}822{,}608 - 13{,}000) \times 1.05$
$= £6{,}116{,}949$ (3)

Liabilities $= 80 \times (10{,}000\ \overset{7.306}{\ddot{a}_{80}}) \times 1.02$
$= £6{,}124{,}896$ (4)

Loss = (4) − (3)
$= £7{,}947$

(e) Interest = 5·3% actual
against 4% expected } Good

Mortality = 3 deaths actual
against ($83q_{79} = 3.92$) deaths expected } Bad

Expenses = £13,000 actual
against (2% × 10,000 + 83 = £16,600 expected) } Good

So the loss is because the low mortality was more significant than the high interest and the low expenses,